

Chapter 4 - Data Analysis and Interpretation

Preliminary Guide-Review Version

Status: The tables use 22 eligible responses from the first Google Forms export. They demonstrate the analysis method and must be recalculated after full data collection.

4.1 Response Screening

The export contained 24 records. Two respondents selected No to the hybrid-training eligibility question and were excluded. The remaining 22 eligible records had complete responses to all 26 Likert items.

Data-quality check	Result
Submitted records	24
Ineligible records excluded	2
Eligible records analysed	22
Eligible records with missing Likert responses	0
Exact duplicate records	0
Same-answer patterns across all 26 items	5

Four of the same-answer records selected Strongly Agree throughout and one selected Strongly Disagree throughout. They are retained and treated as a caution rather than deleted.

4.2 Respondent Profile

Age group	Frequency	Percentage
Below 25	0	0.0
25-34	6	27.3
35-44	11	50.0
45-54	5	22.7
55 and above	0	0.0
Total	22	100.0

Length of service	Frequency	Percentage
Below 5 years	2	9.1
5-10 years	7	31.8
11-20 years	10	45.5
Above 20 years	3	13.6
Total	22	100.0

Job level	Frequency	Percentage
Clerical	2	9.1
Officer	14	63.6
Managerial	6	27.3
Other	0	0.0
Total	22	100.0

The profile variables are used only to describe the preliminary sample. They are not predictors in the current model.

4.3 Construct Descriptives and Reliability

Each construct score is an arithmetic mean on the original 1-5 scale. A mean above 3 indicates average agreement.

Construct	Items	Mean	Standard deviation	Cronbach's alpha
Hybrid Learning	7	3.89	1.25	.987
Trainer Competence	6	3.91	1.30	.987
Learner Engagement	6	3.92	1.35	.989
Training Effectiveness	7	3.64	1.40	.990

Learner Engagement had the highest interim mean. Training Effectiveness had the lowest mean but was still above the neutral score of 3. The alpha values are extremely high. They show that responses moved together in the first 22 records, but do not prove full validation of the instrument. The small sample and same-answer patterns can increase alpha.

4.4 Correlation Results

Relationship	Pearson r	Significance	Interpretation
Hybrid Learning and Learner Engagement	.914	p less than .001	Strong positive association
Trainer Competence and Learner Engagement	.902	p less than .001	Strong positive association
Learner Engagement and Training Effectiveness	.795	p less than .001	Strong positive association
Hybrid Learning and Trainer Competence	.983	p less than .001	Extremely high predictor overlap

The three proposed relationships are positive in pairwise analysis. However, Hybrid Learning and Trainer Competence overlap so strongly that their separate effects cannot be estimated reliably from this interim data.

4.5 Regression of Learner Engagement

Model statistic	Value
R squared	.836
Adjusted R squared	.819
F statistic, df 2 and 19	48.366
Model significance	p less than .001
Variance inflation factor for each predictor	29.571

Predictor	B	Standard error	Standardised beta	t	p
Constant	.080	.412	-	.195	.847
Hybrid Learning	.890	.547	.821	1.625	.121
Trainer Competence	.098	.526	.094	.186	.854

The full model is associated with a large share of variation in Learner Engagement. The Variance Inflation Factor is far above the usual caution level. The current responses rate Hybrid Learning and Trainer Competence so similarly that the model cannot identify their unique effects with confidence.

H1 and H2 are therefore not separately confirmed by this interim regression. The bivariate correlations are positive, but final interpretation needs a larger and more varied sample.

4.6 Regression of Training Effectiveness

Model statistic		Value
R squared		.633
Adjusted R squared		.614
F statistic, df 1 and 20		34.422
Model significance		p less than .001

Predictor	B	Standard error	Standardised beta	t	p
Constant	.421	.579	-	.727	.476
Learner Engagement	.821	.140	.795	5.867	p less than .001

Learner Engagement is positively associated with Training Effectiveness in the interim sample. The model is associated with 63.3 per cent of variation in reported Training Effectiveness. It does not prove a causal relationship.

4.7 Interim Hypothesis Record

Hypothesis	Evidence in 22 eligible records	Interim status
H1	Strong positive correlation; unstable separate regression estimate because of multicollinearity	Inconclusive for separate regression interpretation
H2	Strong positive correlation; unstable separate regression estimate because of multicollinearity	Inconclusive for separate regression interpretation
H3	Strong positive correlation and significant simple regression	Preliminary positive association observed

4.8 Interpretation Boundary

The results describe only the first 22 eligible respondents. Small sample size, non-probability sampling, self-report data, same-answer patterns, and high predictor overlap limit interpretation. The tables show how the answers are derived. They are not final dissertation findings.